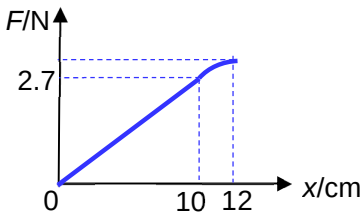


2(a)	$F = kx$ $(0.100)(9.81) = 27x$ $x = 0.0363 \approx 0.04$ x is approximately 4 cm
2(b)(i)	 A graph of Force (F/N) versus extension (x/cm). The y-axis has a tick mark at 2.7. The x-axis has tick marks at 0, 10, and 12. A blue line starts at the origin (0,0) and goes linearly to the point (10, 2.7). From (10, 2.7), the line curves upwards to the point (12, 3.0). Dashed blue lines connect the point (10, 2.7) to both axes, and another dashed line connects (12, 3.0) to the x-axis.
2(b)(ii)	Area under Force-extension graph gives the stored elastic potential energy.
2(b)(iii)	For $x \leq 10$ cm, $E_p = \frac{1}{2} Fx$ $E_p = \frac{1}{2} (kx)x$ $E_p = \frac{1}{2} kx^2$
2(c)	Total extension = $4 + 3 = 7$ cm Increase in E_p due to external force = $E_{p\ 7\text{cm}} - E_{p\ 4\text{cm}}$ $\Delta E_p = \frac{1}{2} (27)(0.07^2) - \frac{1}{2} (27)(0.04^2)$ $\Delta E_p = 0.04455 \approx 0.045 \text{ J}$
2(d)(i)	Removal of external force $\rightarrow$ Stored Elastic potential energy by the force is converted to KE and GPE at equilibrium position $\Delta E_p = KE + GPE$ $0.04455 = \frac{1}{2} mv^2 + mgh$ $0.04455 = \frac{1}{2} (0.100)v^2 + (0.100)(9.81)(0.03)$ $v = 0.549 \approx 0.55 \text{ m s}^{-1}$
2(d)(ii)	As the mass moves upwards, the stored elastic potential energy by the external force is fully converted to kinetic energy and gravitational potential energy as it accelerates upwards towards the equilibrium position. Its kinetic energy increases from 0 J to its maximum at equilibrium position, and gravitational potential energy increases as the mass is displaced vertically upwards. When the mass is moving from equilibrium position to the maximum vertical displacement, its kinetic energy and stored elastic potential energy is converted to gravitational potential energy, and its kinetic energy decreases from a maximum to 0 J, while the elastic potential energy decreases to a minimum.
2(a)	Weight is the force acting on the mass due to the gravitational force of attraction